

when today's prices are high. Past low volatilities and high current prices, therefore, coincide with elevated risk today and in the future.³⁷ At the time of writing, Treasury bonds have record low yields, and so bond prices are very high. Risk-free U.S. Treasuries can be the riskiest investment simply because of high prices. And at a low enough price, risky equities can be the safest investments. Risk parity, poorly implemented, will be pro-cyclical because it ignores valuations, and its pro-cyclicality will manifest over decades because of the slow mean reversion of interest rates.

6. Norway and Wal-Mart Redux

Diversification involves holding many assets. In section 2 we saw that when we started with the United States and progressively added countries to get to the G5 (United States, Japan, United Kingdom, Germany, France), there were tremendous benefits from adding assets. Conversely, if we go backward and remove assets from the G5, we decrease the diversification benefits.

Norway is excluding Wal-Mart on the basis of alleged violations of human rights and other ethical considerations. Removing any asset makes an investor worse off, except when the investor is not holding that asset in the first place. When we are forced to divest an asset, what is the reduction in diversification benefits?

6.1. LOSS OF DIVERSIFICATION BENEFITS

When I teach my case study on Norway and its disinvestment of Wal-Mart in my MBA Asset Management class, I ask the students to compute the lost diversification benefits from throwing out Wal-Mart. We can do this using mean-variance investing concepts. I will not do the same exercise as I give my students, but I will go through an experiment that removes various sectors from a world portfolio. This is also relevant to Norway because as of January 2010, Norway no longer holds any tobacco stocks. Other prominent funds including CalPERS and CalSTRS are also tobacco-free.

I take the FTSE All World portfolio as at the end of June 2012, when it had thirty-nine sectors and 2,871 stocks. What happens if we eliminate tobacco? Let's use mean-variance concepts to quantify the loss of diversification benefits. In this exercise, I compute variances and correlations using a Bayesian shrinkage estimator operating on CAPM betas (see Ledoit and Wolf (2003)) and estimate expected

³⁷ A contrary opinion is Asness, Frazzini, and Pedersen (2012). They argue that investors are averse to leverage and this causes safe assets to have higher risk-adjusted returns than riskier assets. Risk parity allows some investors to exploit this risk premium.